

KHILOK, Russian Federation

WMO: 308440

Lat: 51.367N

Lon: 110.467E

Elev: 2630

StdP: 13.35

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.8	-33.5	-41.8	0.5	-37.3	-37.4	0.7	-33.2	27.7	2.4	25.0	3.6	1.0	90	0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	85.6	63.3	81.7	62.0	78.1	60.7	66.7	79.1	64.8	76.0	63.1	73.3	6.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	94.9	70.7	61.0	88.5	68.7	59.3	83.0	67.4	33.0	78.8	31.4	76.4	30.0	72.8	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.8	20.8	17.8	DB	-44.3	92.3	3.2	4.2	-46.6	95.4	-48.4	97.8	-50.2	100.2	-52.5	103.3	(4)
			WB	-44.0	70.2	3.7	2.2	-46.6	71.7	-48.7	73.0	-50.8	74.2	-53.5	75.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.8	-10.3	-2.2	15.5	33.6	45.6	58.3	64.0	58.7	46.3	30.0	9.5	-5.7
	DBStd	27.12	10.42	10.57	11.41	8.48	7.22	6.63	5.40	6.13	7.22	8.73	11.29	11.57
	HDD50	8805	1869	1461	1071	493	174	11	0	8	156	621	1215	1726
	HDD65	13300	2334	1881	1536	941	600	214	83	208	561	1086	1665	2191
	CDD50	1059	0	0	0	2	39	261	434	278	45	0	0	0
	CDD65	78	0	0	0	0	0	14	51	13	0	0	0	0
	CDH74	1743	0	0	0	2	57	561	807	280	36	0	0	0
	CDH80	490	0	0	0	0	9	161	254	61	5	0	0	0
Wind	WSAvg	6.2	5.3	5.6	6.6	8.1	8.1	6.3	5.5	5.4	6.2	6.2	6.2	5.1
Precipitation	PrecAvg	15.0	0.2	0.2	0.3	0.6	1.0	2.4	3.9	3.2	1.8	0.7	0.4	0.4
	PrecMax	21.9	0.5	0.5	0.9	1.5	2.5	4.4	8.0	7.3	4.1	1.9	0.9	1.4
	PrecMin	8.2	0.0	0.0	0.0	0.0	0.2	0.2	1.2	0.7	0.0	0.0	0.1	0.0
	PrecStd	3.5	0.1	0.1	0.2	0.4	0.6	1.3	1.7	1.5	1.1	0.5	0.2	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	33.5	51.9	68.4	81.2	91.1	91.5	86.2	79.3	63.0	41.2	25.4
		MCWB	20.1	26.6	38.6	47.3	55.7	63.0	66.3	63.9	57.8	46.5	33.2	22.3
	2%	DB	15.6	26.7	44.5	61.0	75.1	84.7	86.5	80.6	71.8	55.1	35.2	20.3
		MCWB	13.4	22.1	33.8	43.6	52.7	60.5	65.3	62.6	55.1	42.3	29.6	18.1
	5%	DB	11.0	21.9	39.4	55.3	70.0	80.8	82.4	76.7	66.6	49.6	30.9	16.9
		MCWB	9.3	18.3	30.6	40.3	49.8	59.6	64.3	62.0	52.3	39.3	26.9	15.0
	10%	DB	6.8	17.5	34.5	50.1	64.8	76.6	78.4	72.7	61.4	44.1	27.2	12.8
		MCWB	5.5	14.8	27.9	37.9	47.4	58.6	62.8	60.4	50.1	36.1	24.1	11.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	26.8	39.0	48.1	57.2	67.1	70.1	67.5	59.6	47.1	33.7	22.3
		MCDB	22.2	32.6	50.9	65.9	78.0	78.7	83.0	79.0	73.3	61.5	40.9	24.9
	2%	WB	13.6	22.2	34.5	44.3	53.9	64.0	67.8	64.9	56.7	42.8	30.1	18.4
		MCDB	15.4	26.3	43.6	58.9	71.7	78.2	81.1	75.7	67.9	53.6	34.5	20.4
	5%	WB	9.4	18.4	31.1	41.2	51.2	61.7	66.1	63.2	54.1	39.6	27.1	15.0
		MCDB	11.0	21.8	38.8	53.9	67.5	74.9	78.2	73.1	63.2	48.1	30.6	16.9
	10%	WB	5.5	14.7	28.2	38.3	48.6	59.9	64.4	61.6	51.3	36.9	24.2	11.3
		MCDB	6.8	17.4	34.4	48.9	62.3	72.4	74.7	70.6	60.1	43.5	27.1	12.7
Mean Daily Temperature Range	5% DB	MDBR	23.9	31.1	29.6	24.4	27.8	27.7	24.5	23.7	25.2	22.9	22.2	21.0
		MCDBR	23.8	35.1	34.1	35.8	41.8	39.7	34.5	32.7	36.8	33.1	24.4	19.1
		MCWBR	22.1	30.4	23.6	19.9	21.2	18.0	15.6	16.3	19.9	20.4	19.3	17.5
	5% WB	MCDBR	23.4	34.0	32.4	32.6	36.7	30.9	27.2	27.0	31.2	30.6	23.3	19.2
		MCWBR	21.8	29.7	22.5	18.6	19.3	14.9	13.4	14.5	17.5	19.1	18.6	17.6
Clear-Sky Solar Irradiance	taub	0.259	0.283	0.317	0.371	0.459	0.449	0.410	0.374	0.335	0.301	0.281	0.263	
	taud	2.190	2.172	2.199	2.218	1.987	2.130	2.310	2.394	2.448	2.459	2.302	2.169	
	Ebn at Noon	239	266	278	274	253	257	266	271	271	261	230	214	
	Edh at Noon	22	30	36	40	53	47	38	34	28	23	20	19	
All-Sky Solar Radiation	RadAvg	469	828	1309	1626	1811	1893	1757	1501	1171	764	453	348	
	RadStd	17	25	47	128	112	124	146	95	95	56	36	26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page